AERO-DASH 2026

Glider Design & Flight Competition

Organized by ***The Robotronics Club, IIT Mandi***

Prologue: The Kamand Valley Origin Story

It started, as most good ideas at IIT Mandi do, with someone getting bored during a lecture and folding a paper airplane. It sailed clean across the hall, caught what can only be described as an unreasonably generous gust of mountain thermal, and refused to land for a suspiciously long time. The professor blamed the wind. The wind blamed the altitude. The altitude, being 760 metres above sea level and thoroughly unbothered, blamed gravity for showing up late to work.

That one paper airplane somehow turned into a group chat, the group chat turned into a whiteboard full of half-serious wing sketches, and the whiteboard turned into the Robotronics Club deciding that if freshers were going to be launching things across rooms anyway, it may as well be organized, timed, and scored. And so, for the very first time, **Aero-Dash** takes flight: part physics lecture, part talent show, part "why-is-my-glider-doing-a-backflip" support group.

And because a first edition should never be too easy, it is not enough for your glider to simply fly – it must fly *and* carry a little extra baggage, quite literally, in the form of heavy metal nuts. Because if there's one thing the mountains of Himachal have taught this campus, it's that everything here eventually has to carry more weight than it was built for.

Overheard at Orientation

"Lift is just gravity's way of saying 'try again in five minutes'."
— Anonymous '29 Batch Student, somewhere in IIT Bombay Techfest

1 Event Overview

Aero-Dash 2026 is a competitive design and engineering event intended for first-year undergraduate students. The competition provides a structured, hands-on (and occasionally hands-scorched-by-hot-glue) introduction to the core principles of aerodynamics, structural design, and iterative engineering practice. Participants will be required to design, construct, and fly unpowered gliders using a set of standardized, lightweight materials provided by the organizing committee – and then, in a plot twist nobody asked for, make that same glider carry a payload of heavy nuts without falling out of the sky like a confused pigeon.

1.1 Objectives

- To develop participants' understanding of fundamental aerodynamic concepts, including lift, drag, thrust, weight, and stability.
- To require participants to evaluate and apply design trade-offs between gliders optimized for **maximum distance and air time** (Round 1) and gliders optimized to **carry max-**

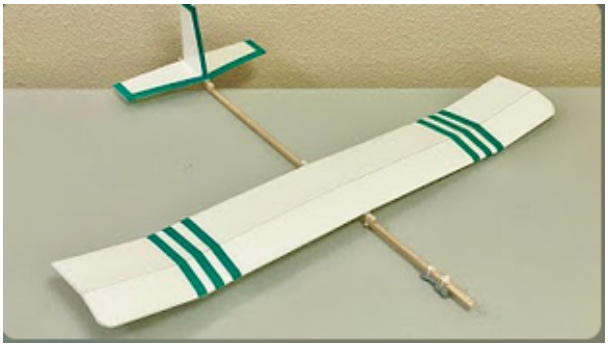


Figure 1: Hehe make me if u can

imum payload while staying airborne (Round 2) – because a glider that only works empty-handed is, frankly, a disappointment.

- ☐ To cultivate teamwork, technical communication, hands-on fabrication skills, and rapid prototyping capability among participants, ideally without anyone gluing their fingers to the fuselage.

2 Learning Resources

Before Phase 1, participants are strongly encouraged (read: expected, unless you enjoy learning aerodynamics the hard way, i.e. by crashing) to review the following video series on glider design fundamentals:

Pre-Event Tutorial Playlist

"Design Your Own Airplanes" – a 10-video playlist covering the basics of glider design, wing selection, and stability.

https://youtube.com/playlist?list=PLpdZAunkELHiN85JpFReHTf-5X-gh6MhV

Watching all 10 videos will not guarantee your glider flies. It will, however, guarantee you sound very confident while it crashes.

3 Competition Timeline and Structure

The competition will be conducted over a single day and organized into five sequential phases, culminating in **two distinct flight rounds**.

Head Judge's Pep Talk

"Time flies. So should your glider."

4 Technical Rules and Specifications

4.1 Dimensional Constraints

All entries must satisfy the following geometric requirements prior to being cleared for flight:

1. **Maximum Wingspan:** 50.0 cm
2. **Maximum Fuselage Length:** 50.0 cm

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Phase 1: Orientation	45 Minutes	Introductory session on aerodynamic principles, launch safety protocol, and a live demo build by the organizers – confidence not guaranteed.
Phase 2: Fabrication	2.5 Hours	Design and construction of entries by participating teams in the designated workshop area.
Phase 3: Trial Testing	30 Minutes	Supervised test flights and adjustment of center of gravity (CG), preferably before something snaps.
Phase 4: Round 1 – No-Payload Trials	45 Minutes	Formal distance and airtime evaluation flown *without* any payload. Two attempts per team.
Phase 5: Round 2 – Payload ("Nut Ladder") Trials	1.25 Hours	Formal payload evaluation using standardized heavy nuts. Two attempts per team.

Table 1: Competition Schedule Overview

There is no minimum or maximum weight restriction on the glider itself; teams may design as light or as sturdy as they like within the wingspan and fuselage length limits above. Do, however, remember that in Round 2 your glider will be asked to politely carry some hardware around the room – a featherweight masterpiece that collapses under one nut will look very elegant and score very little.

Entries that do not conform to the dimensional specifications above will not be eligible for competition and will be disqualified at inspection.

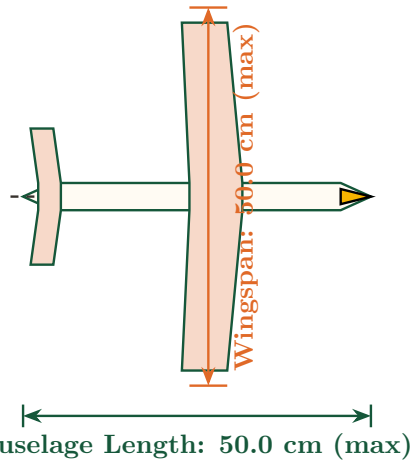


Figure 2: Indicative top-view layout of a competition glider, illustrating the maximum permitted wingspan and fuselage length. Actual planform, taper, and tail configuration are left to each team's design discretion, provided all dimensional and material constraints are satisfied.

4.2 Material Regulations

Approved Fabrication Materials

- ✓ Corrugated plastic sheets (2 sheets provided per team by the organizing committee).
- ✓ Cyanoacrylate (super glue), hot-melt adhesive, or PVA wood glue.
- ✓ Modeling clay or small metal paperclips, restricted to COM Balancing use only.

Prohibited Materials

- ✗ Any form of active propulsion, including electric motors, propellers, rubber bands, or CO_2 cartridges.
- ✗ Pre-molded plastic components or commercially manufactured model aircraft parts.
- ✗ Metal rods, lead weights, or hazardous liquid materials. (The only metal you should be adding on purpose is the payload nuts handed to you in Round 2 – see below.)

Fabrication Committee, 2026

"Corrugated plastic: because balsa wood ghosted us at the last supply order, and we're not talking about it."

5 Round 1: No-Payload Trials

Round 1 is the "prove you can fly at all" round – the glider equivalent of showing up to the first day of college without immediately falling down the amphitheatre steps. No payload is attached. Teams get two attempts, and the better of the two is kept.

5.1 Round 1 Scoring Formula

$$S_1 = D + (2.5 \times T) - P_{penalty} \tag{1}$$

Where:

- D = straight-line distance from the point of launch to the first point of touchdown, measured in meters.
- T = total air time, measured in seconds from release to touchdown.
- $P_{penalty}$ = points deducted per the Penalty Schedule in Section 7.1.

6 Round 2: The Payload "Nut Ladder"

This is where things get heavier, literally. Every entry that clears Round 1 moves on to Round 2, where the objective flips from "fly far and fly long" to "fly long *while dragging a small hardware store behind you.*"

How the Nut Ladder Works

- ✓ The organizing committee will supply standardized **heavy nuts** (steel hex nuts, uniform size and mass, announced at the start of Phase 3 so teams can pre-plan their payload mount).

- ✓ Each team begins Round 2 with one nut affixed at the marked payload station on the fuselage, at or near the design CG, and flies an attempt.
- ✓ Nuts are added one at a time before each subsequent attempt, within the two permitted flights, until either the attempts are exhausted or the glider fails to achieve a valid flight (i.e., it does not glide – it just drops, tumbles, or performs an unscheduled disassembly).
- ✓ Teams must incorporate a simple, secure payload mount into their design during Phase 2 (a small pocket, clip, or tape tab), sized to the standard nut.
- ✓ A payload that detaches mid-flight is not counted toward that attempt's carried weight and incurs the penalty listed in Section 7.1.

6.1 Round 2 Scoring Formula

The score for a given Round 2 attempt depends on the total mass of payload successfully carried through to touchdown and the total time spent in the air with that payload:

$$S_2 = (W_{payload} \times 1) + (T \times 3) - P_{penalty} \tag{2}$$

Where:

- $W_{payload}$ = total mass of nuts still attached at touchdown, measured in grams.
- T = total air time for that attempt, measured in seconds.
- $P_{penalty}$ = points deducted per the Penalty Schedule in Section 7.1.

The higher of the team's two Round 2 attempt scores is taken as their final Round 2 score.

Robotronics Club Motto (Unofficial)

"In Round 2, we don't ask 'will it fly?' We ask 'how many nuts until it doesn't?' "

7 Overall Scoring and Ranking

A team's final competition score is the sum of its best Round 1 score and its best Round 2 score:

$$S_{final} = S_1 + S_2 \tag{3}$$

Teams are ranked in descending order of S_{final}. In the event of a tie, the team with the higher $W_{payload}$ carried in Round 2 is ranked higher (because at that point, bragging rights go to whoever carried more hardware into the sky).

7.1 Penalty Schedule

Infraction	**Penalty ($P_{penalty}$)**
Foot fault at launch line	2 Points
Payload nut(s) detach during flight	8 Points
Structural failure in flight (e.g., wing separation)	10 Points
Exit beyond designated flight boundary	Attempt voided

8 Aerodynamic Fundamentals: Quick Reference

Participants are advised to review the following principles during the Phase 1 orientation session (and, ideally, before Round 2, when your CG will suddenly develop opinions of its own).

8.1 Aerofoil Basics

The **aerofoil** is the cross-sectional shape of the wing, and it is the single biggest factor separating "glider" from "expensive paperweight." A few terms worth knowing:

- **Chord line:** the straight line joining the leading edge (front) and trailing edge (back) of the wing section.
- **Camber:** the curvature of the aerofoil relative to the chord line. A cambered (curved) upper surface generally produces more lift at a given speed than a flat one.
- **Angle of Attack (AoA):** the angle between the chord line and the direction of oncoming air. Lift increases with AoA up to a critical point, beyond which the wing stalls.

In simple terms: air moving over the curved/angled upper surface travels differently than air beneath it, creating a pressure difference that generates lift. You do not need a wind tunnel to understand this – you need a flat sheet, a bit of curvature (or a clever bend), and the patience to test-fly it about six times more than you'd like to.

Because this year's kit is corrugated plastic sheet rather than balsa, most teams will end up building **flat-plate or slightly-bent aerofoils** instead of smoothly curved ones – which brings us to a shape worth knowing about:

A Funny Hint, From One Aerodynamicist to Another

*Look up the **Kline-Fogleman (KFM) aerofoil** before you spend three hours trying to heat-bend a curve into corrugated plastic like it owes you money. The KFM aerofoil is just flat sheets stacked with a little "step" near the leading edge – no compound curves, no steam, no tears. It was practically invented for teams working with flat sheet stock and a deadline.*

— A wing that has given up trying to be elegant, and started trying to fly

8.2 Static Longitudinal Stability

For a stable, unpowered glide, the aircraft's **center of gravity (CG)** must be positioned forward of its aerodynamic **center of pressure (CP)**:

$$X_{CG} < X_{CP} \tag{4}$$

Should the glider pitch upward and stall immediately after release, the CG should be moved forward through the addition of nose ballast. Conversely, if the aircraft dives sharply toward the floor, the nose should be lightened, or positive elevator trim should be applied to the horizontal tail. Because every added nut in Round 2 drags the CG toward the payload station, teams should be prepared to re-trim between attempts – your glider's personality *will* change with each additional nut.

8.3 Dihedral Angle and Lateral Stability

Angling the wings slightly upward relative to the horizontal plane (a dihedral angle of approximately $\Gamma \approx 3°$ to $5°$) provides passive roll recovery when the aircraft is disturbed by

cross-currents – or by the mysterious gust of wind that always seems to appear exactly when the judges are watching.

Archimedes, Probably, If He'd Studied at IIT Mandi

"Give me a lever, a fulcrum, and a bag of heavy nuts, and I shall move the scoreboard."

9 Organizer Checklist and Logistics

9.1 Venue Requirements

- **Indoor Arena:** A gymnasium or enclosed auditorium is required to eliminate wind interference (and to keep the mountain breeze from taking unofficial credit for good flights).
- **Floor Markings:** A metric tape measure should be affixed to the floor along the primary flight axis.
- **Safety Zone:** A clear separation must be maintained between spectator seating and the designated launch trajectory.

9.2 Judges and Staff Assignments

- **Head Judge (1):** Responsible for inspecting dimensions of each glider prior to launch, announcing the standard nut mass in Phase 3, and overseeing the Nut Ladder progression in Round 2.
- **Timekeepers (2):** Operate digital stopwatches synchronized at the launch line.
- **Spotters (2):** Positioned along the flight corridor to record the landing location of each entry and confirm payload retention at touchdown.

10 Contact Information

For inquiries regarding this rulebook, material kits, or registration procedures, please contact:

Event Coordinator: The Robotronics Club, IIT Mandi

Email: robotronics@students.iitmandi.ac.in

Venue: IIT Mandi Kamand Campus – Indoor Sports Complex

The Robotronics Club's Send-Off, First Edition

"Fly high, fall soft, and may your CG always be forward of your CP."